DETAILED ARCHITECTURE · VERIFIABLE AGAINST SOURCE

Context-Aware Cross-Slice U-Net

Glioma segmentation on BraTS 2020. A shared 2-D U-Net encodes each of 5 adjacent slices; a Cross-Slice Transformer fuses them at the bottleneck; a U-Net decoder reconstructs the centre slice. Every value is taken directly from `cst/model.py` and `cst/config.py`.

4.33 M params · baseline U-Net (K=1)

6.70 M params · full model (K=5)

width **base=48**

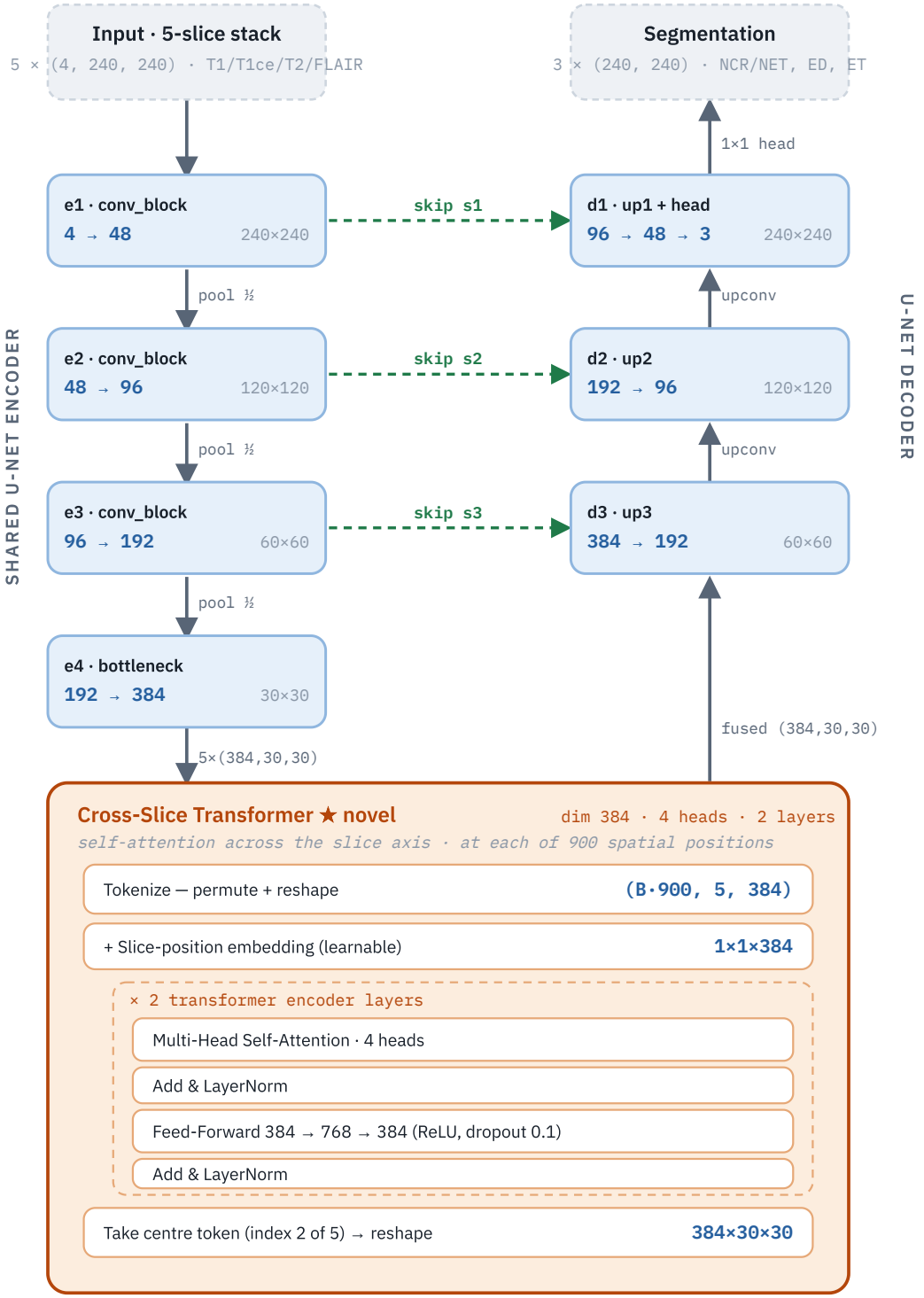
in **B×5×4×240×240**

out **B×3×240×240**

U-Net encoder / decoder (convolutional)

Cross-Slice Transformer — novel module

skip connection (centre slice)



Read it as a “U”: down the left arm (encoder, 240 → 30), across the bottom (cross-slice fusion), up the right arm (decoder, 30 → 240). Dashed lines are the centre slice’s skip connections; **K=1** removes the orange block, leaving a plain U-Net (the baseline).

Layer-by-layer specification `cst/model.py`

`conv_block` = Conv 3×3 (pad 1, bias off) → BatchNorm → ReLU, applied twice.

MODULE	OPERATION	IN → OUT CH	KERNEL	OUTPUT (C×H×W)
ENCODER — SHARED ACROSS ALL 5 SLICES				
e1	conv_block	4 → 48	3×3	48 × 240 × 240 → s1
—	MaxPool2d	48 → 48	2×2 /2	48 × 120 × 120
e2	conv_block	48 → 96	3×3	96 × 120 × 120 → s2
—	MaxPool2d	96 → 96	2×2 /2	96 × 60 × 60
e3	conv_block	96 → 192	3×3	192 × 60 × 60 → s3
—	MaxPool2d	192 → 192	2×2 /2	192 × 30 × 30
e4	conv_block (bottleneck)	192 → 384	3×3	384 × 30 × 30
CROSS-SLICE TRANSFORMER — NOVEL MODULE (ONLY WHEN K > 1)				
tokenize	permute + reshape	384	—	(B·900) × 5 × 384
slice_pos	+ learnable pos-embedding	384	—	(B·900) × 5 × 384
attn ×2	Multi-Head Self-Attention	384, 4 heads	—	(B·900) × 5 × 384
ffn ×2	Linear → ReLU → Linear (+LayerNorm)	384 → 768 → 384	—	(B·900) × 5 × 384
centre	take token index 2, reshape	384	—	384 × 30 × 30
DECODER — RECONSTRUCTS THE CENTRE SLICE (CENTRE-SLICE SKIPS)				
up3	ConvTranspose2d	384 → 192	2×2 /2	192 × 60 × 60
d3	cat(s3) + conv_block	384 → 192	3×3	192 × 60 × 60
up2	ConvTranspose2d	192 → 96	2×2 /2	96 × 120 × 120
d2	cat(s2) + conv_block	192 → 96	3×3	96 × 120 × 120
up1	ConvTranspose2d	96 → 48	2×2 /2	48 × 240 × 240
d1	cat(s1) + conv_block	96 → 48	3×3	48 × 240 × 240
head	Conv2d	48 → 3	1×1	3 × 240 × 240

Hyperparameters & training settings `cst/config.py · train.py`

The values your guide asked about — the current settings a tuning tool (e.g. Optuna) would search over. “Chosen” = validated by the design-justification sweep.

SETTING	SYMBOL / FLAG	VALUE	ROLE
MODEL STRUCTURE			
Slices per stack	`n_slices (K)`	**5**	1 = baseline U-Net · 5 = proposed
Input modalities	`in_modalities`	**4**	T1, T1ce, T2, FLAIR
Output classes	`num_classes`	**3**	NCR/NET, ED, ET
U-Net width	`base_ch`	**48**	channels 48 → 96 → 192 → 384
Attention heads	`xslice_heads`	**4**	multi-head self-attention
Transformer layers	`xslice_layers`	**2**	stacked encoder layers
Feed-forward dim	`dim_feedforward`	**768**	= 2 × 384
Dropout	`dropout`	**0.1**	inside transformer
Image size	`img_size`	**240**	per slice
OPTIMISATION			
Optimizer	`--optimizer`	**AdamW**	chosen (vs Adam / SGD / RMSprop)
Learning rate	`lr`	**2e-4**	chosen (sweep optimum)
Weight decay	`weight_decay`	**1e-5**	regularisation
LR scheduler	`--scheduler`	**cosine**	CosineAnnealingLR (vs step / none)
Loss	`DiceBCELoss`	**Dice+BCE**	bce_weight 0.5, smooth 1.0
Mixed precision	`amp`	**on**	fits 4 GB VRAM
TRAINING REGIME			
Batch size	`batch_size`	**4**	4 GB GPU limit
Epochs	`epochs`	**30**	full runs (config default 80)
Validation split	`val_frac`	**0.20**	by volume/patient (no leakage)
Random seed	`seed`	**42**	reproducibility

Cross-slice attention adapted for multi-modal brain MRI from Abdikenov et al., IEEE Access 2026 (originally CT lymph-node segmentation). All values verifiable in cst/model.py & cst/config.py.